

1 · THE PROBLEM, RESTATED BY THE DATA

The brief says "detect anomalies in video". The dataset says something narrower and harder. Training ships **3,173 short clips of 5–30 s with exactly one event each**, and for 7 of 11 classes the event *is* the whole clip, so there are no usable boundaries. Scoring is **24 short clips (D1, 25 marks), 6 × 240 s (D2, 35) and 4 × 308–629 s (D3, 40)**, where an event counts only at class match **and temporal IoU ≥ 0.5**.

75 of the 100 marks had no training data at all. Every architecture decision below follows from that single fact.

2 · ARCHITECTURE DECISIONS

Windowed VLM, not a frame classifier

20 s windows, 10 s hop, 16 frames per window. Frames are sampled **by timestamp, never by frame index**: clip frame rates run from 1.875 to 30 fps, so index-based sampling silently mis-times every event in the set.
`src/vad/frames.py`

Two output files per run, written separately

Every run emits raw per-window predictions *and* merged events. That separation is why the **72-configuration merge sweep costs zero GPU and no re-inference** — the most productive decision we made, since cheap output policy beat every model change we tried. `src/vad/merge.py`, `sweep.py`

One shared prompt module

Training and inference import the same prompt, asserted byte-identical. If they drift by one token the fine-tune is optimised against a prompt the inference lane never sends — a failure that would show up only as poor accuracy. `src/vad/prompts.py`

The scorer was built before any model

Every result — zero-shot, fine-tuned, ensembled — was routed through one matcher, so runs are comparable to each other and to the portal.
`src/ahc_vad/scoring.py`

A three-state event schema

No row = **not run**; "events": [] = **the model says normal**; "failed": `<error>` = **the run broke here**. A failed video must never be the same bytes as a normal one — conflating them silently inflates every partial run. This distinction caught two of the ten failures in §7.

3 · THE CODEBASE

Two Python packages under `src/`, deliberately separated by owner, plus a Modal driver and the deliverables.

<code>ahc_vad/</code>	taxonomy, Event type, ground-truth and manifest loaders, submission builder and validator, scoring, the synthetic long-video generator, SFT row builder
<code>vad/</code>	the inference lane: prompts, timestamp frame sampling, window planner, an engine abstraction over HF transformers / any OpenAI-compatible server / Gemini native, merge policy, cross-run consensus, runtime instrumentation, the orchestrator and the 72-config sweep
<code>modal_app.py</code>	two Volumes — one for data and outputs, one for the model cache — an A100 inference function, and an <code>sft()</code> that runs a training command verbatim so neither agent second-guesses the other's hyperparameters
<code>demo/ slides/ mv/</code>	the live dashboard and its Modal endpoint, the deck, and the motion-vector study

Ownership was by directory, and cross-boundary changes were requested rather than made. We paid for ignoring that once: one agent rebuilt four modules another had already specified and deleted its own work to resolve it.

4 · EVALUATION METHODOLOGY

The practice pack ships its own answer key

We found `ground_truth.csv` in the downloadable pack, and its event counts match the public leaderboard's denominators exactly. We therefore treated the practice board as **a measuring instrument, never a tuning target**, and said so — a 100.0 on that board is almost certainly the shipped key.

We reverse-engineered the portal metric

From an all-empty submission: D2 returned **11.67** against the portal's 11.7 and D3 **0.00** against 0.0, so **marks = mean per-video credit × max marks**. Reimplemented as `score_per_video()` and it reproduces exactly.

The matcher, and the bug in it

Greedy, **best-IoU-first** — not first-fit. The two diverge whenever several predictions overlap one truth event, which is the common case at D3. First-fit was written, caught in review, and replaced.

Calibration finding that cost us hours. Our strict local matcher scored the D2/D3 output at approximately zero while the portal awarded **38 marks** for the same file. The real metric gives partial credit a strict IoU match does not. We had been under-valuing our own output all afternoon. After the eval pack shipped *without* ground truth, every remaining decision was made from portal feedback, one upload at a time.

5 · MODEL SELECTION

Five models, identical 24-video D1 set, same prompt, same output policy, same A100.

MODEL	P	R	F1	SILENT
Qwen3-VL-4B-Instruct	0.77	0.50	0.61	8/24
Qwen3-VL-8B-Instruct	0.62	0.40	0.48	8/24
Cosmos-Reason2-8B	0.57	0.20	0.30	13/24
Qwen3-VL-2B-Instruct	0.30	0.15	0.20	12/24
Qwen2.5-VL-7B-Instruct	0.00	0.00	0.00	20/24

- **Benchmark leadership does not transfer.** Cosmos-Reason2-8B tops NVIDIA's own Traffic Anomaly Reasoning leaderboard (AI City Challenge 2026, Track 3) and reaches half the recall of a 4B here.
- **Qwen2.5-VL-7B is silent on 20 of 24** — and the mechanism is instruction-following, not blindness: it returns a bare `[]` instead of the requested object. It is ASK-HINT's own published backbone.
- Rejected before testing, with reasons: hosted models (banned from the runtime path), 72B/90B (too large for real time), Llama-3.2-Vision (accepts one image per request; we send 8–32 frames).

We retracted our own headline. "Bigger is worse — the 4B beats the 8B" was published internally, then checked: the two differ on **3 videos out of 24**, sign test **p = 0.25**. A tie, inside the noise band we had been telling everyone to apply. Withdrawn before it reached the deck.

The retraction turned out to be load-bearing, not merely honest. On the private evaluation pack the **8B produced our best submission** — the D1 12.1 and D2 29.8 that make up 41.9 of our 49.9 (it agrees with our best file on 23 of 24 D1+D2 videos, against 21/24 for either 4B run). Had we kept the "bigger is worse" claim we would have shipped the 4B and scored lower. Both models are self-hosted and runtime-legal; only D3 came from a three-model union.

6 · WHAT WE TRIED

Worked — per-difficulty policy

The single best decision, and two uploads isolated it. The difficulties point in **opposite directions**: D2 is 85% correct, so extra coverage **cost 9.5 marks** (29.8 → 20.3); D3 is ~10% correct, so extra coverage **gained 4.0** (4.0 → 8.0). The portal's own advice — cutting false alarms raises marks more than finding extra events — is true in aggregate and **wrong for D3**, where most of the remaining marks sit. **Aggregate guidance hid an opposing signal.**

Worked — cross-run consensus

Keep an event only if ≥2 independent runs agree on the class at overlapping times. On the practice set, where we held the answer key: **precision 0.19 → 0.33, false alarms 56 → 26, recall unchanged**. All 30 dropped predictions were false alarms. This mattered because we had already measured that the model's **own confidence is useless as a filter** — raising the threshold 0.5 → 0.9 removed 2 false alarms out of 31 — so the signal had to come from outside the model's score. `src/vad/ensemble.py`

Failed, each a real negative result

Frame counts 8/16/24/32 · two prompt variants · dense 8 s and 4 s windows · four alternative models · class pruning on the eval set · the LoRA fine-tune (42.4 replacing the baseline, 49.2 added — both below the 49.9 standing at the time) · Qwen3-VL-32B on D3 · targeted per-group prompts · max-recall D1 fill.

Worked — D3 span widening, and the negative result that found it

The negative first. Eight prediction sets — three model sizes, five window configurations, the fine-tune, targeted prompts — **all scored exactly 8.0** on D3. A submission answering only 2 of the 4 videos scored 4.0, so 8.0 was 2.0 per video *answered*: a participation floor, zero events found.

That invariance was the clue. Eight interventions varying the *model* changed nothing, which meant the failure was not in the model. Our D3 spans were **1–8 seconds inside videos of 327–602 seconds** — geometrically incapable of reaching IoU ≥ 0.5 however right the class was. Merging same-class events per video and padding ±60 s gave spans of 112–138 s. **D3 went from 8.0 to a submission scoring 56.0 overall — +6.1, the largest single gain of the day — with no new inference and not one predicted class changed.** Our D3 classes had probably been right all along; we were failing on geometry, not perception.

The evidence was in our own notes from the morning: `.context/01-dataset.md` records a single **125-second** congestion event in the practice pack. We had written down that D3 events are long, then spent hours varying models instead of spans.

7 · TEN SILENT FAILURES

None of them crashed. Each produced a plausible artifact and would have read as "the model just isn't very good".

- `ffmpeg -ss` before `-i` is a keyframe seek and silently returns more than requested — a 468 s video rendered as 241 s.
- 49 training rows with `end ≤ start` — garbage spans, not exceptions.
- Clip pool keyed on **folder name, not class label**, after the organisers relabelled 108 of 164 wrong-way rows mid-event. Would have taught the model that ordinary traffic is an anomaly.
- Empty model output was indistinguishable from a correct "no events" until we recorded the raw text on every empty result.
- A run **exited 0** having completed 6 of 34 videos, writing the other 28 to disk as confident "normal".
- `capture_output=True` buffered a 20-minute A100 job to invisibility.
- **The fix for the fifth caused the seventh**: per-video exception handling turned a loud crash into 10 rows of plausible output and a precision/recall table you could have pasted into a deck.
- Orphaned `ffmpeg` workers wrote into a directory after it was deleted.
- Adding a web endpoint to `modal_app.py` broke every batch job in the file — five queued eval runs and a fine-tune died silently.
- A probe reported **"val acc 1.000"** — meaningless: 77 embeddings, a 7-sample split, a 13 GB upload that had died twice unnoticed.

In a seven-hour build every failure that mattered was silent, so we validated intermediate artifacts rather than outputs. The diagnostic added for the fourth is what caught the seventh.

8 · SMALL DECISIONS THAT EARNED MARKS

- **Top-1 output at D1 cut false alarms 4x (12 → 3) at zero recall cost**, because a D1 video has at most one truth event. Free and CPU-side. Deliberately **not** carried to D2/D3, where it deletes the only true positive.
- `explanation` earns a bonus and can never reduce the score, so we always emit it — **+2.0 of 5** on the practice pack.
- Unanswered videos are scored as normal, so a well-formed **all-empty submission is a valid floor** and the correct first upload to validate the schema end to end.
- Latency is **self-reported**, so we instrumented real p50/p95 instead of quoting a config number: a window closes every 10 s and is scored in **1.83 s — 5.5x real-time headroom, 2.8x at p95**.
- We validated the frame sampler on the awkward clips *before* trusting it: six videos run at 1.875 fps and two hold only 30 frames total, yet 24 requested frames come back unique — **verified by hashing decoded frames, not by counting them**.
- Modal Volumes commit periodically, so we stream child output live. A 20-minute A100 job with buffered output is unobservable.

9 · TWO THINGS WE DID TO OURSELVES ON PURPOSE

We manufactured the missing training data — 235 long multi-event videos, 546 events, with exact ground truth (153 train + 82 dev) — and then **measured the shortcut we had created** rather than assuming it away: concatenation makes event onset ≡ scene cut, so a model learns to detect cuts. After a fix, $P(\text{event} \mid \text{cut})$ 0.10 and $P(\text{cut} \mid \text{event start})$ 0.536. A per-class census showed **6 of 11 classes have zero pre-event footage**, so honest lead-in was impossible for them and interior jump cuts were used instead.